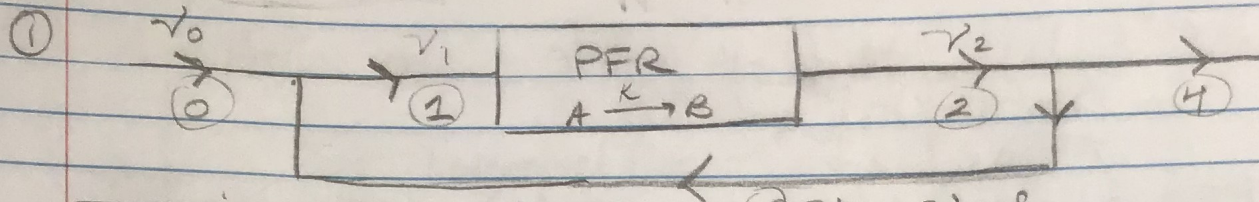


CHE 321 HW3

Aimen
LATEEF
2/12/20



$$X_T = 1 - \frac{C_{A4}}{C_{A0}}$$

total conversion

$$v_3 = v_0 \cdot f$$

$C_{A0}, C_{A1}; C_{A2} = C_{A3} = C_{A4}$ mass bal

$$C_{A2} = C_{A1}(1-X)$$

logically

$$v_1 = v_0(1+f)$$

- PFR -

$$X_{PFR} = \frac{\int FA}{dV} = \tau$$

$$F_{A1} = F_{A0} + F_{A3}$$

$$C_{A3} = C_{A2} = C_{A1}(1-X)$$

$$C_{A0}v_0 + C_{A3}v_3 = C_{A1}v_1 \quad v_3 = v_0 \cdot f$$

$$C_{A0}v_0 + C_{A1}(1-X)(v_0 f) = C_{A1}v_0(1+f)$$

$$C_{A0} + [C_{A1}(1-X)]f = C_{A1}(1+f) - [C_{A1}(1-X)]f$$

$$C_{A0} = C_{A1} + C_{A1}f - C_{A1}f + C_{A1}X$$

$$C_{A0} = C_{A1}(1+fX) \rightarrow C_{A1} = \frac{C_{A0}}{1+fX}$$

mass bal $F_A = F_{A1}(1-X)$

$$r_A = -K C_A = -K(C_{A1}(1-X))$$

$$= -K \left[\frac{C_{A0}(1-X)}{(1+fX)} \right]$$

$$\frac{d[F_{A1}(1-X)]}{dV} = -K \left[\frac{C_{A0}(1-X)}{(1+fX)} \right]$$

$$\frac{d[(1-X) \cdot \frac{C_{A0}}{(1+fX)} \cdot v_0(1+f)]}{dV} = \frac{-K(1-X)}{(1+fX)} \quad \text{where } F_{A1} = C_{A1}v_1 = \left[\frac{C_{A0} \cdot v_0(1+f)}{(1+fX)} \right]$$

$$\frac{v_0(1+f)}{dV} \frac{d \left[\frac{1-X}{1+fX} \right]}{dV} = \frac{-K(1-X)}{(1+fX)} \cdot \frac{1}{v_0(1+f)}$$

left side

$$\frac{(1+fX)d(1-X) - (1-X)d(1+fX)}{(1+fX)^2}$$

implicitly differentiate

$$\frac{dx}{dV} \left(\frac{(1+fX)d(1-X) - (1-X)d(1+fX)}{(1+fX)^2} \right) = \frac{-K}{v_0(1+f)} \frac{dV}{dV}$$

$$\frac{dx}{dV} \left(\frac{1}{(1+fX)^2(1-X)} \right) = \frac{-K}{v_0(1+f)^2}$$

integration

$$\int_0^x \frac{1}{(1+fx)(1-x)} dx = \int \frac{-K}{V_0(1+f)^2} dv$$

PFD → (partial fraction decomposition).

$$\int \frac{1}{(x-1)(fx+1)} dx \rightarrow \int \frac{1}{(f+1)(x-1)} - \frac{f}{(f+1)(fx+1)} dx$$

$$\frac{1}{f+1} \int \frac{1}{x-1} dx - \frac{f}{f+1} \int \frac{1}{fx+1} dx$$

$$\frac{\ln(fx+1) - \ln(1-x)}{f+1} \leftarrow - \left[\frac{\ln(x-1)}{f+1} - \frac{\ln(fx+1)/f}{f+1} \right]$$

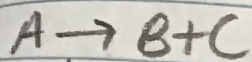
$$\ln \left[\frac{fx+1}{1-x} \right] = -K V \cdot (1+f)$$

$$\frac{1+fx}{1-x} = \frac{K/V_0(1+f)}{e} \rightarrow 1+fx = 1-x \left(e^{\frac{K/V_0(1+f)}{e}} \right)$$

$$X_T = 1 - \frac{C_{A4}}{C_{A0}} = 1 - \frac{F_{A4} \cdot \gamma_4}{F_{A0} \cdot \gamma_0} = 1 - \frac{1-x}{fx+1} \cdot x \left[f + e^{\frac{K/V_0(1+f)}{e}} \right] = \frac{e^{\frac{K/V_0(1+f)}{e}} - 1}{e^{\frac{K/V_0(1+f)}{e}} + f}$$

$$X_T = 1 - \left[\frac{1 - e^{\frac{K \cdot V}{V_0(1+f)} - 1}}{e^{\frac{K \cdot V}{V_0(1+f)}} + f} \right] \left[\frac{f \cdot e^{\frac{K \cdot V}{V_0(1+f)} - 1} + 1}{e^{\frac{K \cdot V}{V_0(1+f)}} + f} \right]$$

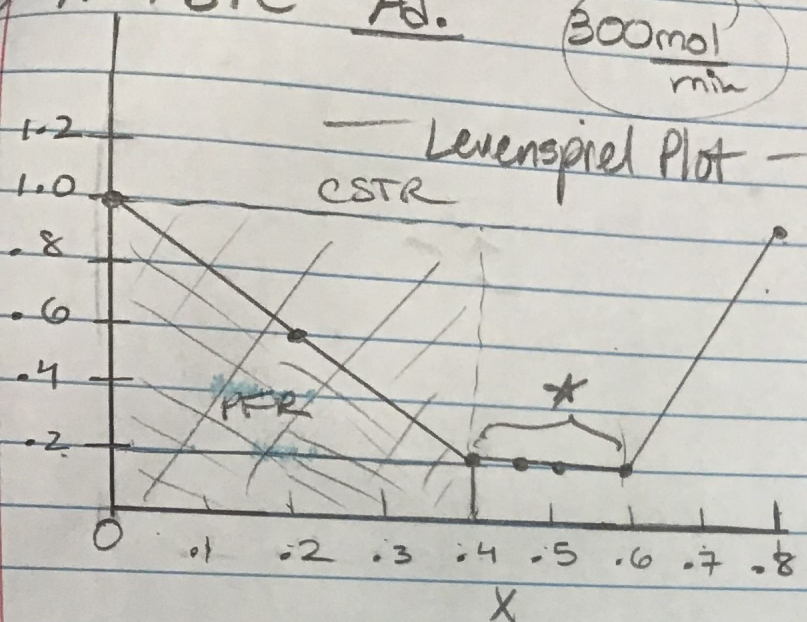
$$x = \frac{e^{\frac{K/V_0(1+f)}{e}} - 1}{e^{\frac{K/V_0(1+f)}{e}} + f}$$



Ad.

$300 \frac{\text{mol}}{\text{min}}$

Amun



X	$-1/r_A$ (min.L/mol)
0	1
.2	.6
.4	.2
.45	.2
.5	.2
.6	.8
.8	1.09

$$\text{PFR} \rightarrow V = \int_0^X \frac{F_{A0}}{-r_A} dX$$

$$\text{CSTR} \rightarrow V = \frac{F_{A0}X}{-r_A}$$

(a) CSTR, PFR vol for 40% conversion?

$$\text{PFR: } (.4 \times .2) + \left(\frac{1}{2} (.4 \times .8) \right) = .24 \frac{\text{min.L}}{\text{mol}} \cdot \frac{300 \text{ mol}}{\text{min}} = 72 \text{ L}$$

$$\text{CSTR: } (.4 \times 2.0) = .8 \frac{\text{min.L}}{\text{mol}} \cdot \frac{300 \text{ mol}}{\text{min}} = 240 \text{ L}$$

(b) Over the range of .4 to .6 conversion the PFR and CSTR volume would be identical, as the integral on this region and "boxed area" would be equal, therefore the volumes would be equal. This is because due to reaction conditions the reaction rate in CSTR is always operating at the lowest rxn rate and the PFR starts high and decreases gradually. This just so happens to be when the conversion is .4-.6.